HANCHU ZHOU

(+1)9167763625 —— zhouhanchu@hotmail.com —— Homepage —— Linkedin

EDUCATION

University of California, Davis, Ph.D. — Sep. 2022 - Present
Electrical and Computer Engineering, Advisor: Junshan Zhang — **Davis, CA**

Zhejiang University, Bachelor — Sep. 2018 - Jun. 2022
Control Science, Advisor: Qinyuan Ren — **Hangzhou, China**

RESEARCH INTERESTS

Robotics; World Model; RL; SFT

SELECTED PUBLICATIONS

CarDreamer: World-Model-Based Autonomous Driving Platform — Apr 2024 - Jul. 2024
IEEE IoT-J: CarDreamer → ★ ***300+ stars***

- Developed the first open-source platform for WM based autonomous driving in response to the lack of an easy-to-use WM based learning platform, with the goal of advancing research in this field.
- Designed various scenarios for RL training with great flexibility on modality, observability, and intention sharing. Trained policy with DreamerV3-based RL from scratch on each scenario.

VITA: Vision-To-Action Flow Matching Policy — May. 2025 - Aug. 2025
ICLR 2026: VITA

- Presented VITA, a novel Vision-To-Action paradigm that treats latent images as the source of the flow, and learns an inherent mapping from vision to action.
- Noise-free VITA outperforms or matches state-of-the-art generative policies, while reducing inference latency by 50% to 130% compared to conventional flow matching policies.

IN-RIL: Interleaved Learning in Robotics Manipulation — Dec. 2024 - May. 2025
arXiv: IN-RIL

- Proposed a new training approach by using Imitation Learning and Reinforcement Learning in an interleaving way to ensure the learning stability and generalization capability in robotics manipulation.
- Conducted extensive experiments on various robotics benchmarks and demonstrated the superior performance of our approach with up to 633% improvement on success rate.

World-Model-Based Hierarchical Planning — Mar 2024 - Jun. 2024
Under review: HANSOME

- Introduced HANSOME, a WM based hierarchical planning with semantic communication framework, aiming to address the challenges posed by partial observability and non-stationary in autonomous driving.
- Proposed AdaSMO to balance the training of higher-level and lower-level policies in an entropy-controlled adapative way. The extensive experiments show that HANSOME outperforms SOTA WM-based methods in challenging driving tasks by 116% and enhances overall traffic safety and efficiency.

EI-Drive: Edge-Intelligent Autonomous Driving Platform — Jun. 2023 - Aug. 2024
IEEE IoT-J: EI-Drive

- Developed a full-stack autonomous driving platform featuring edge communication and cooperative perception, enabling the exploration of V2X communication's impact on autonomous driving performance.

SKILLS

Programming:	Python, C, C++, MATLAB, HTML, Markdown, C#, JavaScript
Machine Learning:	JAX, PyTorch, TensorFlow, Scikit-learn, Pandas, OpenCV, Huggingface
Tools:	Open3D, YOLO, Wandb, Matplotlib, Git, ROS, Linux, Conda